

NoisePrivacyMap<L2Distance<RBig>, PrivacyCurveDP> for ZExpFamily<2>

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This proof resides in “contrib” because it has not completed the vetting process.

Proves soundness of the implementation of `NoisePrivacyMap` for `ZExpFamily<2>` with output measure `PrivacyCurveDP`.

1 Hoare Triple

Precondition

Compiler-verified

NoisePrivacyMap is parameterized as follows:

- MI is `L2Distance<RBig>`.
- MO is `PrivacyCurveDP`.

Caller-verified

None.

Pseudocode

```
1 class ZExpFamily2:
2     def noise_privacy_map(
3         self, _input_metric: L2Distance[RBig], _output_measure: PrivacyCurveDP
4     ) -> PrivacyMap[L2Distance[RBig], PrivacyCurveDP]:
5         scale = self.scale
6         if scale < RBig.ZERO: #
7             raise "scale must be non-negative"
8
9         def privacy_map(d_in: RBig):
10             if d_in < RBig.ZERO: #
11                 raise "sensitivity must be non-negative"
12             if d_in == RBig.ZERO: #
13                 return PrivacyCurve().with_zCDP(0.0)
14             if scale == RBig.ZERO: #
15                 return PrivacyCurve().with_zCDP(float("inf"))
16
17             rho = f64.inf_cast((d_in / scale) ** 2 / RBig(2)) #
18             return PrivacyCurve().with_zCDP(rho) #
19
20         return PrivacyMap.new_fallible(privacy_map)
```

Postcondition

Theorem 1.1. Given a distribution `self`, the implementation returns an error if `self` is invalid. Otherwise it returns a privacy map for the mechanism $\text{function}(\mathbf{x}) = \mathbf{x} + \mathbf{Z}$, where the coordinates of $\mathbf{Z}$ are independent discrete Gaussian samples represented by `self`, under `PrivacyCurveDP`.

Proof. Line 6 rejects a negative scale, which does not represent a valid discrete Gaussian distribution. Line 10 rejects a negative distance bound.

When `d_in` is zero, adjacent inputs are identical and line 12 returns the identity privacy curve. When the sensitivity is positive and the scale is zero, the mechanism has no finite privacy guarantee; line 14 returns the non-private curve represented by $\rho = \infty$.

It remains to consider positive rational sensitivity s and scale σ . By Theorem 14 of [CKS20], the discrete Gaussian mechanism is ρ -zCDP for

$$\rho = \frac{1}{2} \left(\frac{s}{\sigma} \right)^2.$$

Line 17 performs the arithmetic exactly over rationals and then casts toward positive infinity, so the returned floating-point value $\hat{\rho}$ satisfies $\hat{\rho} \geq \rho$.

Line 18 constructs a `PrivacyCurve` from this zCDP guarantee. Its privacy-profile and tradeoff queries are conservative conversions of the same zCDP guarantee. Therefore it is a valid `PrivacyCurveDP` distance for the mechanism. Increasing the distance to any `d_out` accepted by the measure ordering preserves the guarantee, which proves the privacy-map postcondition. $\square$

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.